

Farmalengke — Product Requirements Document

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Draft status. This PRD is a working draft, not a baseline. Four scoping decisions have been confirmed with the client (business model, forward-contract scope, vault custody, target market); everything still marked [NEEDS INPUT] in §14 must be closed before the client can sign Gate 5 and set baseline v1.0.

1. Overview & Objective

Farmalengke is a Philippine agricultural trading platform that connects smallholder farmers to buyers through both a physical consolidation point (the *bagsakan*, or trading port) and a digital equivalent. It operates as a **principal buyer** — purchasing produce outright at agreed prices and carrying inventory risk — and simultaneously as a **marketplace**, where third-party buyers transact directly with farmers for a commission. Around that trading core sit three supporting capabilities: a logistics network of freelance and small-fleet delivery providers, a predictive analytics tool that forecasts yield and price to support planting and forward-contracting decisions, and a Vault that holds prepayment and funding as a Stellar-based stablecoin balance governed by a Soroban smart contract.

Objective: increase the share of the final selling price that reaches the farmer, while giving buyers dependable volume and quality — measured by farmgate price realization and fulfilment reliability, per the KPIs in §5.

2. Background & Context

Philippine agricultural supply chains are characterised by long trader chains between farm and market. Produce typically passes through several intermediaries, each taking margin, and smallholders — who lack storage, transport, price information, and pre-harvest liquidity — have weak bargaining power at the point of sale. The *bagsakan* already exists as an institution: a physical drop-off and consolidation point where farmers deliver and traders buy.

Farmalengke's premise is that digitising the *bagsakan* — rather than replacing it — is the realistic intervention. Farmers keep a physical place to deliver, while the platform adds price transparency, guaranteed offtake, forward commitments, and pre-harvest cash.

Why now. Mobile penetration and digital wallet adoption (GCash, Maya) are now high enough among rural Philippine users to make a digital settlement layer viable, which was not true a decade ago.

Evidence gap. The paragraphs above describe the problem as understood from the client brief. The specific market sizing, current trader-margin percentages, and adoption figures that would justify the investment case are

not yet sourced. See [NEEDS INPUT] Q1 and Q2 in §14 — these must be evidenced before Gate 2, not assumed.

Constraints envelope. Budget: [NEEDS INPUT] Q3 . Timeline: [NEEDS INPUT] Q4 . Technology stack is fixed by the client (§10). Target market is the Philippines first, with architecture prepared for later multi-jurisdiction expansion.

3. Problem Statement

Job to be done — farmer: *"When my crop is ready, I need to convert it to cash quickly and at a fair price, without losing it to spoilage or being forced to accept whatever the buyer at the bagsakan offers me that morning."*

Job to be done — buyer: *"I need a reliable volume of produce at a known quality and a known price, on a date I can plan around."*

Pain points

Who	Pain
Farmer	No visibility into prevailing prices; accepts the price offered on the day
Farmer	No pre-harvest liquidity; borrows from traders at unfavourable terms, often repaid in produce
Farmer	Post-harvest loss when produce cannot be moved or sold in time
Farmer	No forward certainty — plants without knowing whether there will be a buyer
Buyer	Inconsistent volume and quality; no way to plan supply
Buyer	No recourse when a supplier fails to deliver
Logistics	Fragmented, informal, unpredictable trips and payment

Current workarounds. Farmers sell to whichever trader appears; rely on informal credit from those same traders; use text messages and personal networks for price discovery. Buyers maintain their own trader relationships. None of this is recorded, which is why neither side can build a track record that would earn better terms.

4. Users & Personas

Persona	Role / permissions	Volume (target, to be validated — [NEEDS INPUT] Q5)	Goals
Smallholder farmer	Own profile, own listings, own contracts, own Vault balance. Cannot see others' prices beyond published market rates	500 in year 1	Sell at a fair price, get paid fast, access pre-harvest cash

Persona	Role / permissions	Volume (target, to be validated — [NEEDS INPUT] Q5)	Goals
Farmer group / cooperative	All farmer permissions, plus aggregate listing on behalf of member farmers and view of member deliveries	10 groups	Consolidate member volume for better terms
Bagsakan operator	Record intake, grade and weigh produce, confirm receipt, manage physical inventory at one site	2 sites	Move produce through the port accurately and fast
Buyer (wholesaler, institutional, processor)	Browse listings, place orders, enter forward contracts, view own order history	50	Reliable volume, known quality, planned delivery
Logistics provider (freelance rider / small fleet)	Accept delivery jobs, update status, upload proof of delivery, view own earnings	30	Steady jobs and prompt payment
Platform trading desk	Set buy prices, accept/decline produce, manage platform-owned inventory and positions	3 staff	Trade profitably without over-committing capital
Treasury / finance admin	Vault operations, reconciliation, payout approval, cash-out rails. Dual-control on movements	2 staff	Funds correct, reconciled, auditable
Compliance officer	KYC review, AML alerts, audit log access. Read-only on trading	1 staff	Meet regulatory obligations, evidence them
Platform administrator	User management, role assignment, configuration. No unilateral fund movement	2 staff	Keep the platform running safely

Permissions principle. No single role can both initiate and approve a movement of funds. Trading desk sets prices but cannot release Vault funds; treasury releases funds but cannot set prices. This separation is a requirement, not an implementation detail (FR-27, FR-28).

5. Goals & Success Metrics

ID	Goal	Success metric (KPI)
G1	Raise the share of final sale price reaching the farmer	Farmgate price realization rises 15% against a baseline measured in the first 60 days of operation, by month 12
G2	Reduce post-harvest loss on produce moving through the platform	Spoilage and rejection below 5% of transacted volume by weight, by month 12

ID	Goal	Success metric (KPI)
G3	Give buyers dependable supply	95% of confirmed orders fulfilled in full, on the agreed date, at the agreed grade, by month 12
G4	Give farmers pre-harvest liquidity	200 farmers receive at least 1 prepayment; default rate below 8% of disbursed value, by month 12
G5	Improve planting and pricing decisions	Price forecast mean absolute percentage error below 20% at a 14-day horizon, on the top 5 commodities by volume
G6	Establish viable platform unit economics	Platform gross margin per transacted kilo turns positive by month 9, sustained 3 consecutive months
G7	Operate lawfully in the Philippines	Zero regulatory findings; required registrations and licences in place before the first peso of public funds is held

G7 is deliberately a first-class goal, not a compliance footnote. The Vault holds customer money; operating it without the correct licence is an existential risk to the whole platform, not a feature defect. See RISK-1 and RISK-2 in §12.

6. Scope & MVP

MoSCoW

- **Must:** Farmer signup and KYC; produce listing; bagsakan intake with grading and weighing; platform principal purchase; marketplace listing and order placement; order fulfilment and delivery tracking; logistics provider signup and job assignment; Vault balance, prepayment disbursement and settlement; payout to Philippine cash-out rails; audit logging; role-based access control; Filipino and English interface.
- **Should:** Forward contracts between buyer and farmer; farmer credit scoring from delivery history; yield forecasting; price forecasting; cooperative/group accounts; buyer ratings; dispute resolution workflow; offline-tolerant field data capture.
- **Could:** Multi-bagsakan inventory transfer; produce quality photo capture with automated grading assistance; buyer subscriptions/standing orders; SMS-only interaction for farmers without smartphones; logistics route optimisation.
- **Won't (now):** Tradable derivatives or price speculation by non-participants; secondary trading of forward contracts; consumer/retail direct-to-household sales; farm input (seed, fertiliser) marketplace; equipment leasing; crop insurance underwriting; expansion beyond the Philippines.

The MVP boundary. MVP is a *single bagsakan site, single region, five commodities*, operating the principal-buyer model end to end — farmer signs up, lists produce, delivers to the bagsakan, is graded and weighed, is paid from the Vault, and the platform resells to a buyer who ordered through the marketplace. Logistics is included but may be manually dispatched. Analytics ships as descriptive reporting only in MVP; forecasting follows once there is transaction history to train on.

Explicitly out of scope

1. Any instrument tradable by a party who does not intend to take or make delivery. The client confirmed forward contracts are commercial supply agreements, not financial products; the platform must not become a derivatives venue by accident.
2. Secondary-market resale of forward contracts (same reason).
3. Lending to farmers as a standalone product. Prepayment is advance consideration against a specific committed delivery, not a general-purpose loan — this distinction is material to whether a lending licence is required, and must be confirmed by counsel (**[NEEDS INPUT]** Q6).
4. Custody of customer crypto assets other than the single stablecoin used for settlement.
5. Non-Philippine jurisdictions in MVP, though the data model must not preclude them.
6. Native mobile applications in MVP — the platform ships as a responsive web application.

7. Functional Requirements

Each story carries acceptance criteria and the goal (G#) it serves.

7.1 Farmer signup & identity

- **FR-1 (G1)** — **As a** smallholder farmer **I want** to register with my mobile number and a one-time PIN **so that** I can join without an email address or a computer.
- *Acceptance criteria:* Registration completes with mobile number + OTP only; no email required; OTP expires after 5 minutes; 3 failed OTP attempts locks the number for 15 minutes; the flow completes in under 2 minutes on a 3G connection; interface available in Filipino and English.
- **FR-2 (G7)** — **As a** compliance officer **I want** each farmer to complete identity verification proportionate to their transaction value **so that** the platform meets its KYC obligations without excluding farmers who lack formal documents.
- *Acceptance criteria:* Tiered KYC — Tier 1 (name, mobile, barangay, photo) permits transactions up to a configurable ceiling; Tier 2 (government ID + selfie match) required above it; specific thresholds set from the legal opinion in Q6 ; a farmer blocked at a tier boundary sees what document is needed and why; all verification decisions are timestamped and attributable to a named reviewer.
- **FR-3 (G1)** — **As a** farmer **I want** to register my farm — location, area, and the crops I grow **so that** buyers and the platform know what I can supply.
- *Acceptance criteria:* Farm captures GPS point or barangay selection, area with unit (hectare or square metre), and one or more crops with expected harvest months; location capture works offline and syncs later; a farmer may register multiple farms.
- **FR-4 (G1)** — **As a** cooperative officer **I want** to register my group and link member farmers **so that** we can sell consolidated volume.
- *Acceptance criteria:* A group has a named officer; farmers join by invitation and must accept; a member can leave; group listings show constituent member contributions; a farmer's individual identity and Vault balance stay separate from the group's.

7.2 Produce listing & the digital trading port

- **FR-5 (G1)** — **As a** farmer **I want** to list produce I have or expect to have **so that** the platform or a buyer can purchase it.

- *Acceptance criteria:* A listing captures commodity, variety, quantity with unit, quality grade claimed, harvest or availability date, and location; a listing is either *available now* or *expected* with a date; listings can be created offline and sync on reconnection; a farmer can edit or withdraw a listing until it is committed to an order.
- **FR-6 (G1) — As a farmer I want** to see the current platform buying price and recent market prices for my commodity **so that** I can judge whether to sell now.
- *Acceptance criteria:* Price view shows the platform's current bid, the 7-day price range, and the source and timestamp of each figure; no price is displayed without its source; prices older than 48 hours are visibly marked stale.
- **FR-7 (G3) — As a buyer I want** to browse and filter available produce **so that** I can find the volume and quality I need.
- *Acceptance criteria:* Filter by commodity, grade, quantity, location, and availability date; results show quantity, grade, distance, and earliest delivery date; a buyer sees only listings they are eligible to purchase.
- **FR-8 (G3) — As a buyer I want** to place an order against one or more listings **so that** I can secure supply.
- *Acceptance criteria:* An order aggregates one or more listings; placing it reserves the quantity and prevents double-selling; the order states total price, delivery date, and delivery location; the farmer is notified and must confirm within a configurable window (default 24 hours) or the reservation lapses.
- **FR-9 (G6) — As the platform trading desk I want** to set and publish buying prices per commodity and grade **so that** the platform buys at a margin it controls.
- *Acceptance criteria:* Prices set per commodity, grade, and bagsakan site, with an effective-from timestamp; a price change never retroactively alters a committed order; every price change is logged with the user who made it; prices can be scheduled ahead.

7.3 Forward contracts

- **FR-10 (G1, G3) — As a buyer I want** to commit now to purchase a future harvest at an agreed price **so that** I secure supply and the farmer gets certainty.
- *Acceptance criteria:* A contract records commodity, grade, quantity, agreed unit price, delivery window, delivery location, and both parties' acceptance; both parties must accept explicitly; the accepted terms are immutable thereafter and any variation creates a new linked version; the contract is presented in plain Filipino and English before acceptance.
- **FR-11 (G4) — As a farmer I want** part of a forward contract's value advanced to me at signing **so that** I can fund inputs for the season.
- *Acceptance criteria:* Advance is a configurable percentage of contract value, capped per farmer by credit limit (FR-13); advance is disbursed from the Vault only after both parties accept and the buyer's funds are committed; the advance is recorded as a liability against the contract and is netted at settlement.
- **FR-12 (G3) — As a party to a forward contract I want** clearly defined consequences if delivery falls short **so that** I know my exposure before I sign.
- *Acceptance criteria:* Shortfall handling is stated in the contract terms before acceptance; the system computes settlement for full, partial, and zero delivery; a shortfall opens a dispute case (FR-24) rather than silently deducting; force-majeure conditions (typhoon, flood, pest declaration) are recorded as a distinct outcome from ordinary default and do not count toward the farmer's default rate.

Design note. FR-12's force-majeure carve-out matters for G4's default-rate KPI. In a typhoon-exposed country, treating weather loss as farmer default would both misstate the metric and penalise farmers for events outside their control.

7.4 Bagsakan (trading port) operations

- **FR-13 (G2)** — **As a bagsakan operator I want** to record produce arriving at the port against the farmer and the expected listing or contract **so that** intake is attributed correctly.
- *Acceptance criteria:* Intake records farmer, commodity, gross and net weight, grade assessed, arrival timestamp, and operator identity; intake can be matched to an existing listing, order, or contract, or recorded as a walk-in; intake works offline and syncs on reconnection with conflicts surfaced for human resolution, never silently merged.
- **FR-14 (G2)** — **As a bagsakan operator I want** to grade and weigh produce with the farmer present and record their agreement **so that** disputes about grade are prevented rather than argued afterwards.
- *Acceptance criteria:* Grade assessment records the grading standard applied, the grade assigned, the weight, and the operator; the farmer confirms on the device or the disagreement is recorded; a photograph of the delivered lot is captured and stored against the intake record; the farmer receives a receipt by SMS or in-app within 60 seconds of confirmation.
- **FR-15 (G2)** — **As a bagsakan operator I want** to track produce held at the site **so that** we know what is on hand and how long it has been there.
- *Acceptance criteria:* Inventory shows quantity by commodity and grade, age since intake, and assigned order if any; lots exceeding a per-commodity age threshold are flagged; every movement (in, out, transferred, written off) is logged with a reason and a user.
- **FR-16 (G2)** — **As the trading desk I want** to record spoilage and rejection **so that** loss is measured rather than absorbed invisibly.
- *Acceptance criteria:* Write-off records quantity, reason, and responsible stage; write-offs are reportable against G2's 5% target; a write-off above a configurable value requires second-person approval.

7.5 Logistics

- **FR-17 (G3)** — **As a delivery provider I want** to register with my vehicle and capacity **so that** I receive jobs I can actually do.
- *Acceptance criteria:* Registration captures identity, vehicle type, capacity in kilograms, whether refrigerated, licence and insurance where required, and service area; documents are verified before the first job is assigned; expiry dates are tracked and a provider with a lapsed document stops receiving jobs.
- **FR-18 (G3)** — **As a delivery provider I want** to see and accept available jobs **so that** I can plan my day.
- *Acceptance criteria:* Job shows pickup, destination, commodity, weight, required vehicle type, time window, and payment amount before acceptance; acceptance is first-come or assigned per configuration; an accepted job is removed from other providers' views; a provider can decline without penalty before acceptance.
- **FR-19 (G3)** — **As a buyer I want** to see where my delivery is **so that** I can plan receiving.
- *Acceptance criteria:* Status transitions (assigned, collected, in transit, delivered, failed) are visible with timestamps; each transition is attributable to a user or device; location updates where the provider has granted permission; a failed delivery requires a recorded reason.

- **FR-20 (G3)** — **As a delivery provider I want** proof of delivery captured **so that** my payment is not disputed later.
- *Acceptance criteria:* Proof captures recipient name, signature or photograph, and timestamp; proof is captured offline and syncs; payment is released against confirmed proof per FR-26.

7.6 Predictive analytics

- **FR-21 (G5)** — **As a farmer I want** an indicative price forecast for my commodity **so that** I can decide when to sell or whether to commit forward.
- *Acceptance criteria:* Forecast presents a range, not a point estimate; every forecast shows its confidence interval, the data it used, and its recent accuracy; where data is insufficient the tool states that instead of producing a number; forecasts are never presented as advice or a guarantee.
- **FR-22 (G5)** — **As the trading desk I want** yield and volume forecasts by region and commodity **so that** I can plan buying capacity and working capital.
- *Acceptance criteria:* Forecast covers a 30, 60, and 90-day horizon; shows expected volume with interval; identifies the top 3 drivers of each forecast; back-tested accuracy is published alongside and refreshed monthly.
- **FR-23 (G5)** — **As a platform administrator I want** forecast accuracy tracked against outcomes **so that** we know whether the tool is working.
- *Acceptance criteria:* Every forecast is stored with its inputs and later scored against actuals; MAPE is computed per commodity and horizon; accuracy below the G5 threshold of 20% MAPE at 14 days raises an alert and the forecast is withdrawn from the farmer-facing view until it recovers.

Sequencing. FR-21 through FR-23 cannot ship at MVP: there is no transaction history to train on at launch. Analytics ships first as descriptive reporting over actual platform trades, and forecasting activates only once accuracy clears the G5 bar on back-testing. Publishing a forecast that is worse than a farmer's own judgement would actively harm the people this platform exists to serve.

7.7 Vault, payments & settlement

- **FR-24 (G4)** — **As a farmer I want** a Vault balance showing what I am owed, what has been advanced, and what I can withdraw **so that** I understand my position.
- *Acceptance criteria:* Balance distinguishes available, committed/held, and advanced amounts; every movement has a timestamp, counterparty, and reference; balance is displayed in Philippine pesos as the primary unit regardless of the underlying settlement asset; history is exportable.
- **FR-25 (G4)** — **As a buyer I want** my payment held until delivery is confirmed **so that** I am not exposed to non-delivery.
- *Acceptance criteria:* Funds move to a held state on order confirmation; release is triggered only by confirmed delivery (FR-14 or FR-20) or by dispute resolution (FR-29); neither party can unilaterally release held funds; the hold and release are enforced by the Soroban contract, not by application code alone.
- **FR-26 (G4)** — **As a farmer or delivery provider I want** to withdraw my balance to a Philippine account or wallet **so that** the money is usable.
- *Acceptance criteria:* Cash-out supports at least GCash, Maya, and InstaPay/PESONet bank transfer; the fee and the amount received are shown before confirmation; a withdrawal request produces a reference

number; funds arrive within 1 banking day for wallet rails; failures are surfaced with a reason and the balance is restored.

- **FR-27 (G7)** — **As a** treasury administrator **I want** every movement of platform funds to require two named approvers **so that** no individual can move money alone.
- *Acceptance criteria:* Movements above a configurable threshold require initiation and approval by two distinct authenticated users; the same user cannot fill both roles; the requirement is enforced by the Soroban contract's multi-signature configuration, not solely in the application layer; every approval is logged immutably.
- **FR-28 (G7)** — **As a** compliance officer **I want** an immutable audit trail of every fund movement, KYC decision, and price change **so that** we can evidence compliance.
- *Acceptance criteria:* Audit records are append-only and cannot be edited or deleted by any role including administrators; each record carries actor, action, timestamp, and before/after state; the trail is exportable for a named date range; on-chain settlement events are reconcilable to off-chain records with a documented procedure.
- **FR-29 (G3)** — **As** either party to a transaction **I want** to raise a dispute and have held funds frozen until it is resolved **so that** neither side can be defrauded.
- *Acceptance criteria:* Raising a dispute freezes the associated held funds; both parties can submit evidence including photographs; a named platform adjudicator resolves with a recorded reason; resolution outcomes (release, partial release, refund) are executed through the Vault; first response within 2 business days.
- **FR-30 (G7)** — **As a** treasury administrator **I want** daily reconciliation between on-chain Vault state and off-chain ledger **so that** discrepancies are caught within one day rather than at audit.
- *Acceptance criteria:* Reconciliation runs at least once every 24 hours; any discrepancy raises an alert to treasury and compliance within 15 minutes of detection; unreconciled discrepancies block new prepayment disbursement until cleared.

7.8 Cross-cutting workflows and edge cases

- **FR-31 (G1)** — **As a** farmer with a basic phone and no reliable data connection **I want** to transact **so that** connectivity does not exclude me.
- *Acceptance criteria:* Core farmer flows (view price, confirm intake, check balance) are reachable by SMS or USSD, or via an operator acting on the farmer's behalf at the bagsakan with the farmer's recorded consent; the fallback path is tested as a first-class flow, not a degraded one.
- **FR-32 (G2)** — **As the** system **I want** to handle the case where delivered quantity or grade differs from what was ordered **so that** partial deliveries settle correctly.
- *Acceptance criteria:* Settlement recomputes on actual delivered weight and assessed grade; the buyer is notified and must accept the variance or open a dispute; variance beyond a configurable tolerance requires explicit buyer acceptance before release.
- **FR-33 (G7)** — **As the** system **I want** to behave safely when the Stellar network or the stablecoin issuer is unavailable **so that** users are not left in an indeterminate state.
- *Acceptance criteria:* Vault operations fail closed — an operation that cannot be confirmed on-chain is never reported as successful; pending operations are queued and reconciled; users see an honest status rather than an optimistic one; the platform can continue accepting non-financial operations (listing, intake, delivery) while settlement is degraded.

8. Non-Functional Requirements

Targets below are engineering proposals derived from the personas and volumes in §4. They are **our** numbers, not the client's, and must be confirmed at Gate 3 (**[NEEDS INPUT]** Q7).

Area	Requirement (with a number)	Goal
Performance — farmer views	Price view and Vault balance render within 2 seconds on a 3G connection at the 95th percentile	G1, G4
Performance — search	Buyer listing search returns within 1 second at the 95th percentile for up to 10,000 active listings	G3
Performance — intake	Bagsakan intake record saves locally within 500 milliseconds, independent of network state	G2
Payload budget	Farmer-facing initial page load under 300 KB compressed, to keep cost on metered data below 1 peso per session	G1
Availability — trading	99.5% monthly uptime for listing, ordering, and intake, measured excluding scheduled maintenance of at most 4 hours per month	G3
Availability — Vault	99.9% monthly uptime for balance reads; settlement writes may degrade but must fail closed per FR-33	G4, G7
Scale — year 1	500 farmers, 50 buyers, 30 logistics providers, 2 bagsakan sites, 200 transactions per day, sustained	G6
Scale — headroom	Architecture supports 10x year-1 volume (5,000 farmers, 2,000 transactions per day) without redesign	G6
Concurrency	200 concurrent authenticated sessions with no degradation beyond the stated 95th-percentile latency targets	G3
Throughput — settlement	At least 500 Vault settlement operations per hour at peak harvest	G4
Response time — disputes	First human response to a raised dispute within 2 business days	G3
Data durability	Zero tolerance for lost financial records; point-in-time recovery to any moment in the preceding 30 days	G7
Recovery	RPO 15 minutes, RTO 4 hours for the trading platform; RPO 0 for Vault state, which is reconstructable from on-chain history	G7
Security — authentication	Multi-factor required for all staff roles; session timeout 15 minutes for treasury and compliance roles, 12 hours for farmers	G7
Security — authorization	Row-level security enforced in the database for every table holding user data; no table relies on application code alone for access control	G7
Security — encryption	TLS 1.3 in transit; AES-256 at rest; no private key material in application code, environment files, or version control at any time	G7

Area	Requirement (with a number)	Goal
Security — key custody	Vault signing keys held in a hardware security module or equivalent, with 2-of-3 multi-signature for treasury movements	G7
Security — testing	Independent penetration test of the Vault and settlement path passed before holding any public funds; zero unresolved high or critical findings	G7
Privacy / compliance	Philippine Data Privacy Act (RA 10173) compliance including NPC registration; GDPR-equivalent data subject rights implemented from day 1 to avoid rework at expansion	G7
Privacy — data minimisation	Farmer location stored at barangay granularity by default; GPS point captured only with explicit per-farm consent and retained 24 months	G7
AML / financial crime	Transaction monitoring with alerting on configurable thresholds; sanctions screening at onboarding and rescreened at least every 90 days	G7
Accessibility	WCAG 2.2 Level AA for all farmer- and buyer-facing screens; minimum touch target 44 by 44 pixels; usable at 200% zoom	G1
Accessibility — literacy	Farmer-facing copy at a reading level appropriate to 6 years of formal schooling; every monetary action confirmable from an icon and number alone	G1
Localization	Filipino and English at launch, with the string layer supporting at least 3 further languages (Cebuano, Ilocano, Hiligaynon) without code change	G1
Localization — units	Metric primary, with local trade units (kaban, sako) displayed alongside and configurable per commodity per region	G1
Device support	Functional on Android 9 and later, on devices with 2 GB RAM, on a 3G connection	G1
Observability	Structured logging, metrics, and alerting on all financial paths; alert to on-call within 5 minutes of a settlement failure	G7
Auditability	Every financial record retained a minimum of 10 years, per Philippine record-keeping expectations, subject to confirmation in Q6	G7

9. Data Requirements

Entity	Source	Sensitivity / classification	Retention	Owner
Farmer profile	Farmer at signup	Personal / sensitive — name, mobile, address, government ID	Active + 10 years (financial record)	Farmer, processed by platform
KYC documents & selfie	Farmer at verification	Highly sensitive — biometric and identity documents	5 years after relationship ends, then destroyed	Farmer
Farm record	Farmer	Personal — GPS location reveals residence	24 months for GPS; barangay indefinitely	Farmer

Entity	Source	Sensitivity / classification	Retention	Owner
Produce listing	Farmer	Commercial, low	Active + 3 years	Farmer
Intake / grading record	Bagsakan operator	Commercial, medium — basis for payment	10 years	Platform
Order & contract	Buyer + farmer	Commercial, high — legally binding	10 years	Both parties
Forward contract terms	Both parties	Commercial, high — immutable once accepted	10 years	Both parties
Vault ledger entry	Platform + chain	Financial, high	10 years, immutable	Platform (regulated)
On-chain settlement record	Stellar network	Public by nature — must contain no personal data	Permanent, outside platform control	Public ledger
Payout / cash-out record	Payment rail	Financial, high — includes bank/wallet identifiers	10 years	Platform
Logistics provider profile	Provider	Personal — identity, licence, insurance	Active + 5 years	Provider
Delivery record & proof	Provider	Commercial, medium — may contain a signature image	3 years	Platform
Dispute case & evidence	Parties	Commercial + personal	10 years	Platform
Price observation	Platform + external	Public / low	Indefinite	Platform
Forecast & its inputs	Analytics	Commercial, medium	Indefinite (needed for accuracy scoring)	Platform
Audit log	System	High — append-only	10 years, immutable	Platform

Critical data rule. No personal data is ever written on-chain. The Stellar ledger is public and permanent; a national ID number or phone number placed there cannot be retracted and would be an irreversible Data Privacy Act breach. On-chain records carry opaque identifiers only, resolved to people through the off-chain database.

Data ownership. Farmers own their production and transaction history. A farmer who leaves may export it and request erasure of everything not required by financial record-keeping law. This is a product commitment, not merely a legal one — a platform that traps a farmer's track record recreates the dependency it exists to break.

10. Integrations & Dependencies

Fixed technology stack (client-specified, not open for redesign):

Layer	Technology	Notes
Frontend & API	Next.js (App Router) with TypeScript	Responsive web; server components for low-bandwidth rendering
Database, auth, storage	Supabase, self-hosted	Self-hosting is a client requirement — it implies the team owns Postgres operations, backups, and upgrades
Vault settlement	Stellar, with a Soroban smart contract	Holds escrow, enforces multi-signature release
Settlement asset	A Stellar-issued stablecoin	Specific asset and issuer [NEEDS INPUT] Q8

External integrations

Integration	Purpose	Risk if unavailable
GCash / Maya	Farmer and provider cash-out	Farmers cannot access funds — critical
InstaPay / PESONet	Bank transfers	Degrades to wallet-only cash-out
SMS gateway	OTP, receipts, notifications for basic phones	Signup and FR-31 blocked — critical
Stellar network + Horizon/RPC	Settlement	Fail closed per FR-33
Stablecoin issuer	Value stability, redemption	Single point of failure — see RISK-3
Weather data service	Yield forecasting inputs	Forecast quality degrades; not critical at MVP
Market price sources (DA, trading posts)	Price benchmarks for FR-6	Price display loses its comparison baseline
Sanctions / PEP screening	AML obligation	Onboarding blocked for compliance reasons

Dependencies and migration. There is no existing system to migrate from; the repository currently holds only scaffolding. Where farmers or bagsakan operators keep paper or spreadsheet records, a one-off import path is a *Should*, not a *Must*.

11. UX / Design & Brand

Design references and brand. **[NEEDS INPUT]** Q9 — no brand guidelines, logo, or palette exist yet. If none are supplied, the recommendation is to run the brand guidelines process before UI build begins, so the interface is not styled twice.

Design principles, which follow from the users rather than from taste:

1. **Money is always legible.** Amounts in pesos, large, with the direction of movement unmistakable. A farmer must never be uncertain whether they are receiving or paying.

2. **Nothing hidden behind literacy.** Every critical action is expressible in an icon and a number. Text supports the interface; it is not the only route through it.
3. **The network is assumed hostile.** Every screen has a defined state for offline, slow, and stale data. "Loading" forever is a defect.
4. **Irreversible actions look irreversible.** Accepting a forward contract or confirming a grade is visually distinct from browsing, and states the consequence before the tap.
5. **The physical and the digital agree.** What the bagsakan operator's screen says must match the paper receipt in the farmer's hand, or trust in the platform collapses.

Accessibility target. WCAG 2.2 Level AA, tested on a low-end Android device on a throttled 3G connection — not only in a desktop browser.

12. Assumptions, Constraints & Risks

Assumptions (each is a belief that, if wrong, changes the plan)

1. Farmers in the target region have access to an Android smartphone, their own or a household member's. **If wrong**, FR-31's SMS/USSD path becomes a *Must* at MVP and the scope grows materially.
2. A licensed cash-out partner will onboard the platform. **If wrong**, farmers cannot convert Vault balances to spendable money and the product does not function.
3. The platform can obtain the licences implied by holding customer funds. **If wrong**, the Vault must be redesigned around a licensed third party — see RISK-1.
4. Bagsakan operators can be trained to grade consistently. **If wrong**, G2 and G3 are both unreachable, since every settlement rests on the grade.
5. Sufficient transaction history to train forecasts accumulates within 12 months. **If wrong**, FR-21 through FR-23 slip beyond year 1.

Constraints

- Technology stack is fixed (\$10), including self-hosted Supabase.
- Philippine market first; architecture must not preclude other jurisdictions.
- Budget and timeline: **[NEEDS INPUT]** Q3, Q4 .
- The platform carries inventory and price risk as principal buyer, so working capital is a hard operating constraint, not only a funding question.

Risk log

#	Risk	Impact	Likelihood	Mitigation	Owner
RISK-1	Holding customer funds as a platform-custodied stablecoin likely requires BSP Virtual Asset Service Provider registration and possibly e-money authority. Operating without it risks cease-and-desist and personal liability	H — existential	H	Obtain written legal opinion before any code touches customer funds (Q6); design the Vault so a licensed third-party custodian can be substituted without redesigning the trading platform	Client sponsor

#	Risk	Impact	Likelihood	Mitigation	Owner
RISK-2	Prepayment against future harvest may be characterised as lending, triggering a separate licence	H	M	Same legal opinion (Q6); structure advances strictly as consideration under a delivery contract, never as standalone credit	Client sponsor
RISK-3	Stablecoin issuer failure, depeg, or freeze would strand farmer balances in an asset they never chose to hold	H	M	Hold the shortest possible duration in the asset; sweep to fiat on a defined schedule; select an issuer with attested reserves; define an exit procedure before launch	Treasury
RISK-4	Farmers bear currency and technology risk they did not consent to and may not understand	H — ethical	M	Denominate and display everything in pesos; the stablecoin is settlement plumbing the farmer never has to reason about; disclose in plain Filipino	Product
RISK-5	Self-hosted Supabase makes the team responsible for Postgres operations, backup, patching, and uptime	M	H	Budget dedicated operations capacity from day 1; document runbooks; test restore monthly, not merely configure backups	Engineering
RISK-6	Principal-buyer model consumes working capital and concentrates price risk on the platform	H	H	Position limits per commodity; start marketplace-heavy and grow the principal book only as capital allows; daily mark-to-market	Trading desk
RISK-7	Forward contracts could be construed as regulated derivatives if any resale or speculation emerges	H	L	Enforce non-transferability in the contract and in code; monitor for proxy trading; §6 out-of-scope items 1 and 2 are load-bearing controls, not preferences	Compliance
RISK-8	Grading disputes at the bagsakan undermine trust in the platform	M	H	Photograph every lot; farmer confirmation at grading (FR-14); published grading standard; dispute path (FR-29)	Operations
RISK-9	Forecasts that are wrong could cause farmers real financial loss if acted upon	H — ethical	M	Never publish a forecast that fails the G5 accuracy bar (FR-23); always show intervals and accuracy	Product

#	Risk	Impact	Likelihood	Mitigation	Owner
				history; never phrase output as advice	
RISK-10	Smart contract defect could lock or lose funds irreversibly	H	M	Independent audit before mainnet; testnet soak; upgrade/pause mechanism with multi-signature governance; cap total value held during the initial period	Engineering
RISK-11	Two-sided marketplace cold start — neither farmers nor buyers join first	H	H	Principal-buyer model directly mitigates this: the platform is the guaranteed first buyer, so farmers need not wait for demand to appear	Commercial
RISK-12	Platform competes with existing traders who may respond by undercutting or pressuring farmers	M	M	Focus on the services traders do not offer (forward certainty, recorded history, transparent grading); consider partnering with traders as buyers rather than displacing them	Commercial

13. Timeline & Milestones

Durations are **relative** — no absolute dates can be set until the budget and start date in Q3 and Q4 are known. Sequencing, however, is not arbitrary: the gates below are ordered by dependency, and M0 genuinely blocks everything financial.

Milestone	Content	Depends on
M0 — Legal & compliance gate	Written legal opinion on custody, prepayment, and forward contracts (Q6); licence path agreed; stablecoin and cash-out partners selected	Nothing — starts immediately, runs parallel to M1
M1 — Foundation	Repository, environments, self-hosted Supabase, schema with row-level security, auth, roles, audit logging, CI	—
M2 — Farmer & listing	FR-1 to FR-9: signup, KYC tiers, farm records, listings, price display, buyer browse and order	M1
M3 — Bagsakan operations	FR-13 to FR-16: intake, grading, inventory, write-off, offline capture and sync	M1, M2
M4 — Vault (testnet)	FR-24 to FR-30 on Stellar testnet: Soroban contract, escrow, multi-signature, reconciliation. No real funds	M1, M0
M5 — Logistics	FR-17 to FR-20: provider onboarding, job assignment, tracking, proof of delivery	M2, M3

Milestone	Content	Depends on
M6 — Security & audit	Independent smart-contract audit; penetration test; remediation to zero high and critical findings	M4, M5
M7 — Pilot (one bagsakan)	Live with a limited farmer cohort, real money, capped exposure. Forward contracts (FR-10 to FR-12) enabled here	M0 licences in hand , M6
M8 — Analytics	FR-21 to FR-23: descriptive reporting first; forecasting activates only on passing the G5 accuracy bar	6+ months of M7 transaction history
M9 — Scale	Second bagsakan site, wider commodity range, cooperative accounts	M7 proven against G1, G2, G3

The critical path runs through M0, not through engineering. Building the Vault before the legal position is settled risks writing a system that cannot lawfully be operated.

14. Open Questions

#	Question	Owner	Blocks
Q1	[NEEDS INPUT] What is the current trader margin between farmgate and wholesale for the target commodities, and what is the source? G1's "15% improvement" needs a measured baseline	Client	Gate 2
Q2	[NEEDS INPUT] Market sizing: how many farmers, what volume, in the target region?	Client	Gate 2
Q3	[NEEDS INPUT] Budget envelope, including the working capital required for the principal-buyer book	Client	Gate 4
Q4	[NEEDS INPUT] Target launch date and any fixed external deadline (grant, season, investor)	Client	Gate 4
Q5	[NEEDS INPUT] Are the year-1 volumes in §4 the client's actual targets, or placeholders?	Client	Gate 3
Q6	[NEEDS INPUT] Legal opinion — does platform-custodied stablecoin require BSP VASP registration or e-money authority? Is prepayment lending? Are forward contracts commercial supply or regulated instruments? What are the record-retention obligations?	Client + counsel	M0, and all of M4/M7
Q7	[NEEDS INPUT] Confirm or revise the §8 non-functional targets — they are our proposals, not the client's	Client	Gate 3
Q8	[NEEDS INPUT] Which stablecoin and issuer? What is the redemption path to pesos?	Client + treasury	M4
Q9	[NEEDS INPUT] Brand guidelines, logo, palette — do these exist, or should they be created first?	Client	M2
Q10	[NEEDS INPUT] Which region and which 5 commodities for the MVP?	Client	Gate 3, M2

#	Question	Owner	Blocks
Q11	[NEEDS INPUT] Which grading standard applies per commodity — a DA standard, a trade convention, or platform-defined?	Client + operations	M3
Q12	[NEEDS INPUT] Who is the named approver with authority to baseline this PRD, and who signs each gate?	Client	Gate 1
Q13	[NEEDS INPUT] Does the platform hold produce on consignment as well as buying outright? This changes revenue recognition and risk	Client	Gate 3
Q14	[NEEDS INPUT] What happens to a farmer's advance if the harvest fails entirely through no fault of theirs? Confirm the policy behind FR-12's force-majeure handling	Client	Gate 3
Q15	[NEEDS INPUT] Is there an existing bagsakan site, operator relationship, or buyer commitment to build the pilot around?	Client	M7

15. Approvals

See the sign-off log (`signoff-log-farmalengke.md`). Baseline v1.0 is set at the Final PRD approval; change control applies thereafter.

This PRD is not yet baselined. Gate 1 cannot be signed until Q12 names the approver, and Gate 5 cannot be reached while Q6 — the legal opinion governing whether the Vault can lawfully operate as designed — remains open.